

BMI Calculator App

A Modern Flutter, Riverpod & SharedPreferences Application

Developer & Architecture Deck

Executive Summary

Product Vision

- Simplify Health Tracking: Empower users to monitor their BMI easily using standard height & weight dimensions.
- Interactive Design: Leverage sliders & steppers for clean, fluid input operations.
- Offline-First Caching: Ensure absolute convenience by persisting data directly to user's storage.

Core Value Proposition

- Dynamic WHO Thresholds: Real-time visual feedback with orange, green, and red alerts corresponding to weight category.
- Zero Latency database interactions: Operations execute instantly on local hardware without cloud delays.
- Premium Visuals: Clean Material 3 cards, rounded corners, clear spacing, and modern typography hierarchy.

Technical Stack Overview

Frontend & UI

- Flutter Framework: Enables cross-platform rendering (iOS, Android, Web, Windows).
- Dart Language: Strong types and null-safety ensure runtime stability.
- Material 3 Design: Built with card elevations, seed colors, and modern rounded buttons.

State & Logic

- Riverpod State: Replaces traditional Provider for safer, compile-time checked state.
- Separation of Concerns: Separate models keep logic completely decoupled from view layer.
- BMINotifier: Central engine handling weight/height steps, BMI formula, and history dispatching.

Local Storage

- SharedPreferences: Asynchronous persistence API.
- JSON Serializer: Encodes health models into JSON strings for local memory block saving.
- Reverse Chronology: Injected items automatically prepended to display latest logs first.

Core App Features

1

Splash & Route

Loads application, pre-fetches local database cache, and redirects to home workspace in 2 seconds.

2

Dynamic Input

Utilizes slider bars for height values (120-220 cm) and incremental weight steppers for instant adjustment.

3

Logic Engine

State changes fire the BMI formula, categorizing health status into Underweight, Normal, Overweight, or Obese classes.

4

History log

Maintains persistent records. Supports clearing records via a modal confirmation alert dialog.

Codebase Architecture & File Map

Data Models & Core Logic Providers

- lib/main.dart - Setup global ProviderScope
- lib/model/bmi_model.dart - Immutable BMI state
- lib/model/history_model.dart - Serialization models
- lib/provider/bmi_provider.dart - Inputs & calculation logic
- lib/provider/history_provider.dart - Local history CRUD notifier
- lib/services/shared_pref_service.dart - SharedPreferences API

UI Pages & Custom Widgets

- lib/screens/splash_screen.dart - Animated splash screen
- lib/screens/main_screen.dart - Bottom navigation bar setup
- lib/screens/home_screen.dart - UI workspace layouts
- lib/screens/history_screen.dart - History listing & delete options
- lib/widgets/height_card.dart & weight_card.dart - Slider/stepper custom UI
- lib/widgets/result_card.dart & history card.dart - Status card widgets

Future Roadmap

1. Multi-Device Sync

- Cloud Firestore database support integration.
- Firebase User Authentication (Google/Email signup).
- Backup weight logs securely in the cloud database.

2. Data Visualization

- Implement weekly and monthly line charts.
- Visualize health trends over time.
- Integrate flutter fl_chart package to draw curves.

3. Advanced Metrics

- Add Basal Metabolic Rate (BMR) parameters.
- Add body fat percentage algorithms.
- Include Dark Theme toggling controls.